# The proposed entropy conclusion and an open problem

**Abstract**

A complete proof of the proposed assertion is not known in the stated generality. In fact, the assertion would settle the open problem of vanishing Lyapunov exponents for cohomologically rigid diffeomorphisms. We give a complete proof of this reduction, explaining why the usual structural results about globally hypoelliptic vector fields do not supply the requested conclusion.

## 1 Status of the question

The hypothesis in the question is called *global hypoellipticity*. The established theory relates this property to cohomology freeness and gives unique ergodicity with a smooth invariant probability density; see Forni [1, Proposition 2.4 and Theorem 3.10]. It does not, however, supply the proposed general entropy conclusion.

The relevant unresolved issue is explicitly recorded by Karaliolios [3, Introduction, p. 6]: it is unknown whether cohomological rigidity of a diffeomorphism implies that all its Lyapunov exponents vanish. The suspension construction below shows that a proof of the assertion in the question would imply precisely this vanishing result.

Thus the following is a rigorous reduction to an open problem, rather than a purported proof of the requested assertion.

## 2 A sufficient condition for global hypoellipticity

We use smooth positive densities, so that orientability is unnecessary. If $\mu$ is such a density, scalar distributions are identified with continuous linear functionals on $C^\infty(M)$ through the pairing

$$\langle u, \phi \rangle_\mu = \langle u, \phi\mu \rangle.$$

For a vector field preserving $\mu$, the distributional derivative satisfies

$$\langle Xu, \phi \rangle_\mu = -\langle u, X\phi \rangle_\mu.$$

**Lemma 1.** *Let $Y$ be a smooth vector field on a compact manifold $P$, preserving a smooth positive probability density $\mu$. Suppose that for every $f \in C^\infty(P)$ there is $v \in C^\infty(P)$ such that*

$$Yv = f - \int_P f \, d\mu. \tag{1}$$

*Then $Y$ satisfies the global hypoellipticity hypothesis of the question.*

*Proof.* Suppose that $Yu = f$ for $u \in \mathcal{D}'(P)$ and $f \in C^\infty(P)$. Invariance of $\mu$ gives

$$\int_P f \, d\mu = \langle Yu, 1 \rangle_\mu = -\langle u, Y1 \rangle_\mu = 0.$$

Consequently, (1) supplies a smooth function $v$ with $Yv = f$. Set $w = u - v$, so that $Yw = 0$.

For any $\phi \in C^\infty(P)$, choose $a \in C^\infty(P)$ satisfying

$$Ya = \phi - \int_P \phi \, d\mu.$$

Then

$$\langle w, \phi \rangle_\mu = \left( \int_P \phi \, d\mu \right) \langle w, 1 \rangle_\mu + \langle w, Ya \rangle_\mu = \left( \int_P \phi \, d\mu \right) \langle w, 1 \rangle_\mu.$$

Thus $w$ is a constant distribution. It follows that $u = v + w$ is smooth. □

## 3 Suspending a cohomologically rigid diffeomorphism

**Definition 2.** Let $F : N \to N$ be a smooth diffeomorphism of a compact connected manifold, preserving a smooth positive probability density $\nu$. We call $F$ *cohomologically rigid* if, for every $a \in C^\infty(N)$ with $\int_N a \, d\nu = 0$, there is $v \in C^\infty(N)$ satisfying

$$v \circ F - v = a. \tag{2}$$

**Lemma 3.** *The constant-roof suspension of a cohomologically rigid diffeomorphism has a globally hypoelliptic generating vector field.*

*Proof.* Let

$$P_F = (N \times \mathbb{R})/\!\sim, \qquad (x, s+1) \sim (Fx, s).$$

The equivalence relation is generated by the free, properly discontinuous smooth $\mathbb{Z}$-action

$$k \cdot (x, s) = (F^k x, s - k).$$

Hence $P_F$ is a smooth manifold. It is compact because the image of $N \times [0, 1]$ is all of $P_F$, and it is connected because $N$ is connected.

Translation in the second coordinate descends to the smooth flow

$$G_t[x, s] = [x, s + t],$$

whose generating vector field $Y$ is induced by $\partial_s$. Since $F$ preserves $\nu$, the product density $d\nu \, ds$ descends to a smooth positive $G_t$-invariant probability density $\mu$ on $P_F$.

We verify (1). Given $f \in C^\infty(P_F)$, denote its lift to $N \times \mathbb{R}$ by $\widetilde{f}$. Thus

$$\widetilde{f}(x, s+1) = \widetilde{f}(Fx, s).$$

Set

$$c = \int_{P_F} f \, d\mu, \qquad a(x) = \int_0^1 \big( \widetilde{f}(x, r) - c \big) \, dr.$$

By the definition of the suspension measure, $\int_N a \, d\nu = 0$. Cohomological rigidity supplies $v \in C^\infty(N)$ with $v \circ F - v = a$.

Define, for all $s \in \mathbb{R}$,

$$\widetilde{u}(x, s) = v(x) + \int_0^s \big( \widetilde{f}(x, r) - c \big) \, dr.$$

This is smooth. Moreover,

$$\frac{\partial}{\partial s} \big( \widetilde{u}(x, s+1) - \widetilde{u}(Fx, s) \big) = 0,$$

and at $s = 0$ the expression in parentheses equals

$$v(x) + a(x) - v(Fx) = 0.$$

Therefore

$$\widetilde{u}(x, s+1) = \widetilde{u}(Fx, s).$$

It follows that $\widetilde{u}$ descends to a smooth function $u$ on $P_F$, and

$$Yu = f - c = f - \int_{P_F} f \, d\mu.$$

Lemma 1 now proves global hypoellipticity. □

# 4 What the requested assertion would imply

**Proposition 4.** *If the assertion in the question holds for all compact connected smooth manifolds, then every cohomologically rigid smooth diffeomorphism preserving a smooth positive probability density has zero topological entropy and all its Lyapunov exponents vanish almost everywhere.*

*Proof.* Let $F : N \to N$ be such a diffeomorphism and form the suspension from Lemma 3. Its generating vector field satisfies exactly the hypothesis of the question.

The embedded section

$$S = \{[x, 0] : x \in N\} \subset P_F$$

is invariant under the time-one map, since

$$G_1[x, 0] = [x, 1] = [Fx, 0].$$

Thus $G_1|_S$ is conjugate to $F$. Topological entropy cannot increase upon restriction to a compact invariant subset: this follows directly from the definition using separated sets, since every separated set in the subset is also separated in the ambient space. Hence

$$h_{\mathrm{top}}(F) \le h_{\mathrm{top}}(G_1). \tag{3}$$

The entropy of a flow is, by convention, that of its time-one map. Consequently, the requested assertion would give $h_{\mathrm{top}}(G_1) = 0$, and (3) would imply $h_{\mathrm{top}}(F) = 0$.

Metric entropy is bounded above by topological entropy [2], so

$$h_\nu(F) = 0.$$

Let $\lambda_1(x), \ldots, \lambda_d(x)$ denote the Lyapunov exponents, listed with multiplicity. Since $F$ is smooth and $\nu$ is a smooth invariant probability density, Pesin's entropy formula applies [4]:

$$h_\nu(F) = \int_N \sum_{j=1}^{d} \max\{\lambda_j(x), 0\} \, d\nu(x).$$

The integrand is nonnegative. Its integral being zero, every Lyapunov exponent is nonpositive at $\nu$-almost every point.

Finally, choose a Riemannian volume density $m$ and write $d\nu = \rho \, dm$, where $\rho$ is smooth and everywhere positive. Invariance of $\nu$ gives

$$\left|\det D_x F^n\right| = \frac{\rho(x)}{\rho(F^n x)}.$$

Because $\rho$ is bounded above and bounded away from zero,

$$\sum_{j=1}^{d} \lambda_j(x) = \lim_{n\to\infty} \frac{1}{n} \log\bigl|\det D_x F^n\bigr| = 0$$

at every Lyapunov-regular point. A finite collection of nonpositive numbers with sum zero consists entirely of zeros. Thus all Lyapunov exponents vanish $\nu$-almost everywhere. □

*Remark* (The essential gap in a tempting entropy argument). Preservation of a smooth volume gives $\sum_j \lambda_j = 0$, not $\sum_j \max\{\lambda_j, 0\} = 0$. The latter is the quantity occurring in Pesin's entropy formula. Consequently, replacing entropy by the integral of the logarithm of the full Jacobian determinant would be incorrect.

## 5 Conclusion

Proposition 4 shows that the question includes the Lyapunov-exponent vanishing problem explicitly identified as open in [3]. The established global-hypoellipticity results provide cohomological rigidity and a smooth invariant measure, but not the additional vanishing needed for entropy. Therefore a complete proof of the proposed general assertion cannot presently be supplied as an established mathematical result; doing so would require resolving this open problem.

## References


[1] G. Forni, *On the Greenfield–Wallach and Katok conjectures in dimension three*, in *Geometric and Probabilistic Structures in Dynamics*, Contemporary Mathematics **469**, American Mathematical Society, 2008, pp. 197–213. See also arXiv:0706.3981, especially Proposition 2.4 and Theorem 3.10.

[2] L. W. Goodwyn, *Topological entropy bounds measure-theoretic entropy*, Proceedings of the American Mathematical Society **23** (1969), 679–688.

[3] N. Karaliolios, *Cohomological rigidity and the Anosov–Katok construction*, arXiv:1711.02732. See the Introduction, p. 6, for the open question concerning vanishing Lyapunov exponents.

[4] Ya. B. Pesin, *Characteristic Lyapunov exponents and smooth ergodic theory*, Russian Mathematical Surveys **32** (1977), no. 4, 55–114.